

CSC3043S Assignment 1: Language Model Training

Name: Chandik Naidoo

Date: 22/08/2026

Introduction:

This report covers a byte-level BPE tokenizer and modern Transformer language model trained on the TinyStories V2 corpus. The BPE tokenizer follows a GPT-2 pre-tokenization regex with additive merge counting to allow for efficient training on the entire corpus. The model also uses pre-norm Transformer blocks with RMSNorm, SwiGLU, RoPE, and QK-norm; before being trained with AdamW, linear warmup + cosine decay, gradient clipping, and bf16 mixed precision.

Important implementation decisions included a KV cache used for efficient generation and a data loader for streaming the tokenized data without loading the full corpus into RAM. Experiments were then performed including a learning-rate sweep, ablations (RMSNorm, RoPE) and a vocabulary-size study which finally culminated in a 20 000-step final model evaluated on a “held-out” test set.

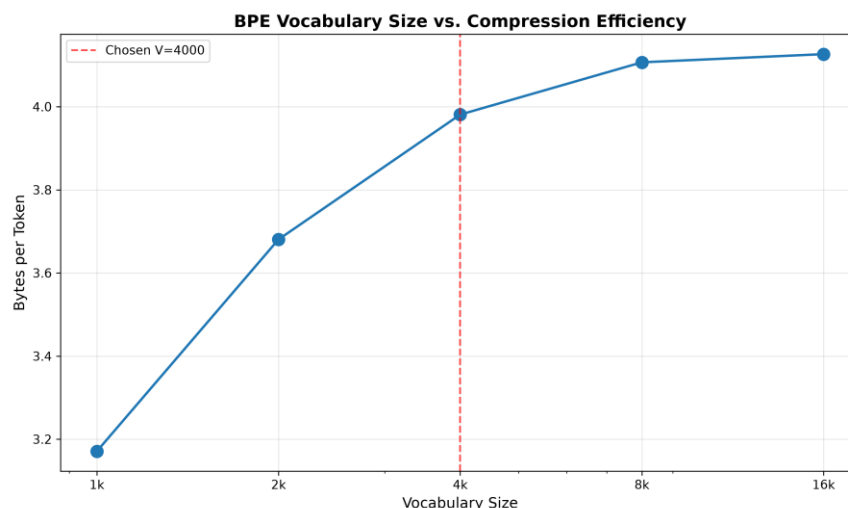
Question 1: Tokenizer Training Time

The V=4000 BPE tokenizer trained on the full 2.2GB TinyStories training file in **239.9 seconds** and encoding the full training file with the same tokenizer took approximately 11 minutes at a throughput of 3.45MB/s (measured on a 100MB slice); both were run on the CPU of a Google Colab instance.

Question 2: Speedup Comparison

Implementation	Time (10 MB slice)/s	Speedup
Tutorial (naïve recount)	348.58	1x
Optimised (incremental)	219.99	1.6x

Question 3: Vocabulary Size Choice



The compression ratio saturates after $V=4000$, where $V=4000$ achieves 3.98 bytes/token, while $V=8000$ only reaches 4.11, which is only 3.3% more than $V=4000$ despite the double in vocabulary size. Also, $V=4000$ uses 4.1M parameters in the embedding and LM head while $V=16000$ uses 16.4M parameters for only a small compression gain of 4.13; therefore $V=4000$ was chosen as the best trade-off between token efficiency and model capacity.

Question 4: Longest Token and Merges

Longest Token: *uncomfortable* (14 bytes, vocab ID – 3237)

First five merges

Merge	Pair (hex)	Decoded
1	20 74	space + 't'
2	68 65	'h' + 'e' → "he"
3	20 61	space + 'a'
4	20 73	space + 's'
5	20 77	space + 'w'

Last five merges

Merge	Pair (hex)	Decoded
1	2064697368 776173686572	"dish" + "washer" → "dishwasher"
2	7265 616479	"re" + "ady" → "ready"
3	206e6577 7370	"new" + "sp" → "newsp"
4	207468696e 6b696e67	"thin" + "king" → "thinking"
5	20616e 63686f72	"an" + "chor" → "anchor"

Yes, the progression makes sense; early merges capture frequent space + letter and digraph pairs, while later merges build whole morphemes, reflecting the expected BPE progression from character combinations to meaningful units.

Question 5: Parameter Count

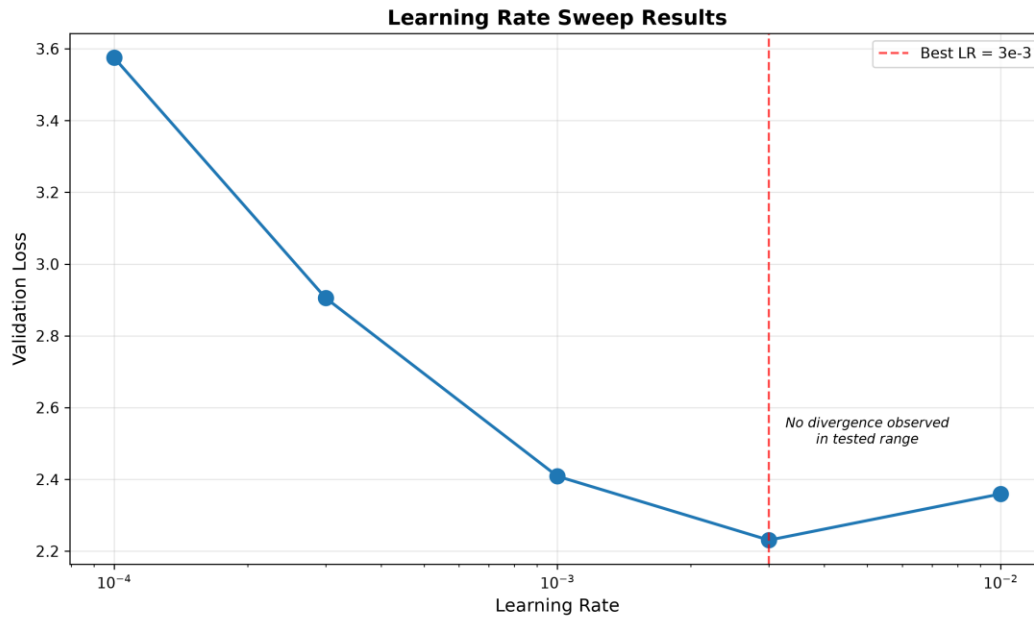
Component	Parameters
Token Embedding	2 048 000
LM Head	2 048 000
Non-embedding	12 456 960
Total	16 552 960

*Also, SwiGLU and ReLU FFN variants do match in total parameters (12 456 960)

Question 6: KV Cache Correctness

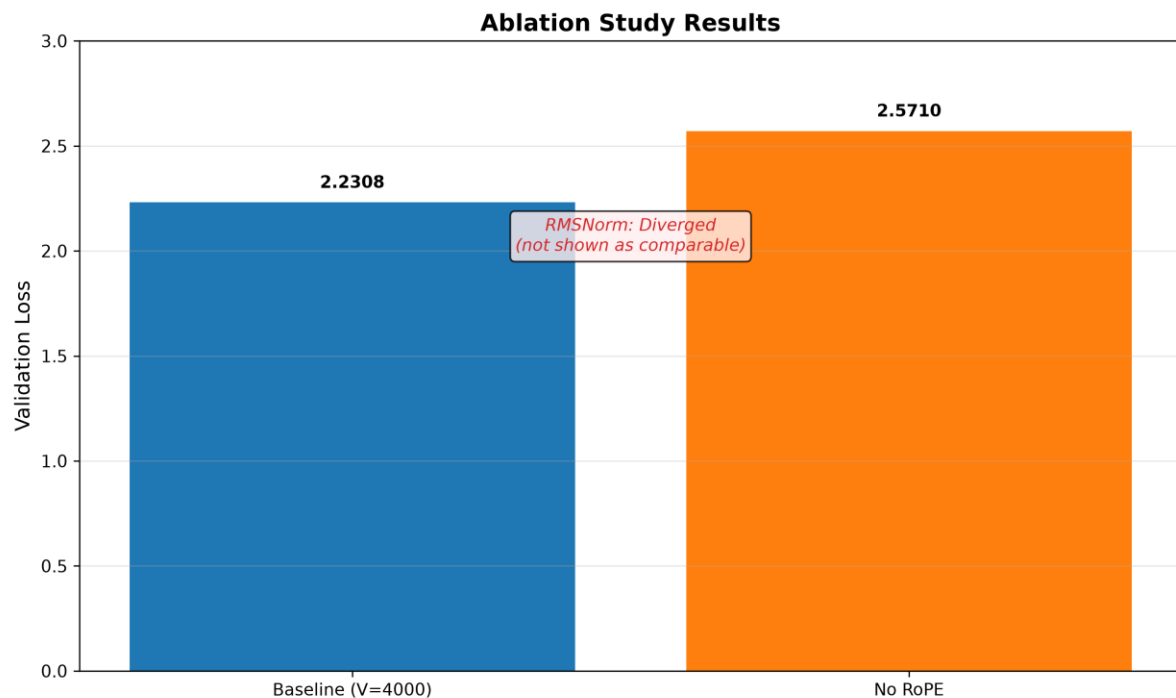
The maximum absolute logit difference between cached and un-cached forward passes was $1.64e-06$. Greedy generations from both paths are bitwise identical over 100 tokens, confirming the KV cache correctly replicates the full forward pass.

Question 10: LR Sweep



The learning rate sweep shows a clear optimum of $3\text{e-}3$ (validation loss: 2.2308). None of the tested rates fully diverged, though $1\text{e-}2$ showed instability.

Question 11 + 12: RMSNorm & NoPE Ablation



The no-RMSNorm model diverged at $\text{LR}=3\text{e-}3$, with gradient norm exploding from 12.47 at step 50 to infinity by step 70. A reduced learning rate was not tested due to GPU budget constraints, but RMSNorm appears critical for training stability at this scale, without it the activation magnitudes are free to grow unchecked across layers.

Removing RoPE increases validation loss from 2.2308 to 2.5710, a gap of 0.3402. This is a meaningful difference as it suggests that even with causal attention, explicit positional encoding via RoPE provides a non-trivial benefit.

Question 14: Design Choice Impact

Choice	Gap	Impact (Relative)
LR (3e-3 vs 1e-3)	0.1780	--
RoPE vs NoPE	0.3402	--
RMSNorm	Diverged	Critical (not numerically comparable)

Among the two measurable ablations, RoPE removal produced the larger loss gap (0.3402 vs. 0.1780 for the LR comparison), suggesting positional information matters more at this scale than the tested learning-rate difference. RMSNorm's removal was however, categorically more severe than either; making it arguably the most critical design choice overall, even though its impact isn't directly comparable on the same loss-gap scale.

Question 15: Vocabulary Size Comparison

Model	Loss	PPL	BPC	Total Params	Embedding + Head
V=1000	1.8838	6.58	0.857	13 480 960	1 024 000
V=4000	2.2308	9.31	0.809	16 552 960	4 096 000

V=4000 achieves lower BPC (0.809 vs. 0.857) despite higher raw loss and PPL (9.31 vs. 6.58), however PPL is not comparable across vocabularies since it counts a different number of prediction events per document, while BPC normalizes by character count and is the fair metric here. V=4000's BPC advantage comes at the cost of 3 072 000 more parameters, concentrated entirely in the embedding and LM head. The trade-off is favourable: a ~22.8% increase in parameters buys a ~5.6% improvement in BPC (from 0.857 to 0.809), making V=4000 the better model for this task despite its higher loss.

Question 18: Final Model

Dataset	Loss	PPL	BPC	Tokens
Validation Holdout	1.5444	4.6851	0.5598	5 185 140
Test Set	1.5618	4.7675	0.5661	404 244

The final model achieves 0.5661 BPC on the held-out test set. This small gap between validation (0.5598) and test (0.5661) BPC suggests the model generalises similarly to unseen data.

Data Split Note: A dedicated test split of the last 2000 validation documents was created after the LR sweep and ablation experiments had been run. Those experiments were evaluated on the full validation file (including what became the test documents). The final model results reported here use the properly split test set, touched exactly once. The validation holdout consists of the remaining documents (first ~25 000 docs).

Question 19: Generated Samples

From the final model (prompt: "Once upon a time,"):

Greedy ($T=0$):

"Once upon a time, there was a little girl named Lily. She had a big, red ball. She loved to play with her ball in the park. One day, she saw a boy named Tom. Tom was sad. Lily wanted to help Tom. Lily said, 'Tom, let's play together!' Tom smiled and said, 'Yes, let's play!' They played catch with the ball. They threw the ball back and forth. They had lots of fun."

Top-p ($p=0.9$, $T=0.8$):

"Once upon a time, there was a little girl named Lily. She lived in a small house with her mom and dad. Lily loved to play with her ball and have fun. One day, Lily was playing with her ball in her room. Her mom said, 'Lily, I want to go to the bedroom. It is so beautiful.' Lily took her mom's ball and went to the park."

Combined ($k=40$, $p=0.9$, $T=0.8$):

"Once upon a time, there was a little girl. She lived in a small town. She had a big, shiny ball. She loved to play with it. One day, she went to the park and saw a new friend. They played together all day long."

I would ship the Combined ($k=40$, $p=0.9$, $T=0.8$) setting: it produces the most coherent, fluent narrative with natural dialogue and no repetition, even though it is quite shorter than the other samples.

Data Availability Note

Several caveats regarding data and reproducibility should be noted:

Compute environment: All GPU experiments were run on free-tier Google Colab (T4 GPU) rather than the assigned Dataspaces L4 allocation, due to persistent access issues with Dataspaces. This is disclosed throughout the report where relevant.

Reduced step count: The LR sweep, ablations, and vocabulary-size runs were conducted at 500 steps rather than the spec's standard 5,000. This reduction was necessary due to GPU budget constraints; the final model was however, trained at the full 20,000 steps (4× standard) as specified.

Late test split: A dedicated test split of the last 2,000 validation documents was created after the LR sweep and ablation experiments had been run. Those experiments were evaluated on the full validation file (including what became the test documents). The final model results in Q18 use the properly split test set, touched exactly once.

Lost per-step logs: The per-step training logs (JSONL) for the LR sweep and ablation runs were lost when the Colab session disconnected before being committed to the repository. All final validation losses reported are from printed outputs during the actual runs, but step-by-step learning curves are not available. The figures in this report show final validation values only, not full training trajectories.

Colab disconnections: The final model training was interrupted multiple times by Colab session disconnections. Resume logic was used to recover from the latest checkpoint (checkpoint_step_8000.pt) and continue training to 20,000 steps.

AI Usage Declaration

Claude (Anthropic) was used as a conversational assistant for debugging, code review, and guidance on experimental design. AGENTS.md was in context for all conversations. All code, experimental design, analysis, and interpretation of results are my own work.